Graphical User Interface (GUI)

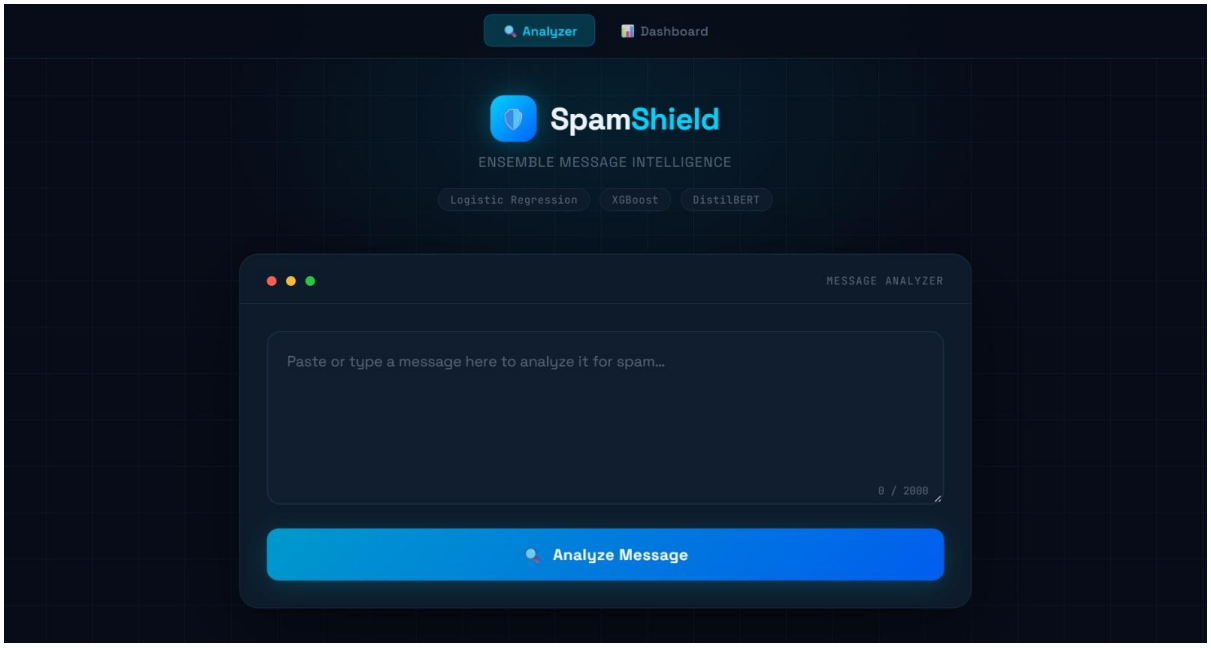


Figure 1. Main SMS Classification Interface.

In the developed system, a Flask technology was used to build a graphical user interface. which provides users with the ability to classify SMS messages in real time. Users can input their SMS messages in the text input box and submit it to get it analysed. As illustrated in Figure 1, the graphical user interface was made simple and easy to use to allow even the users who lack technical knowledge to use the spam classification system.

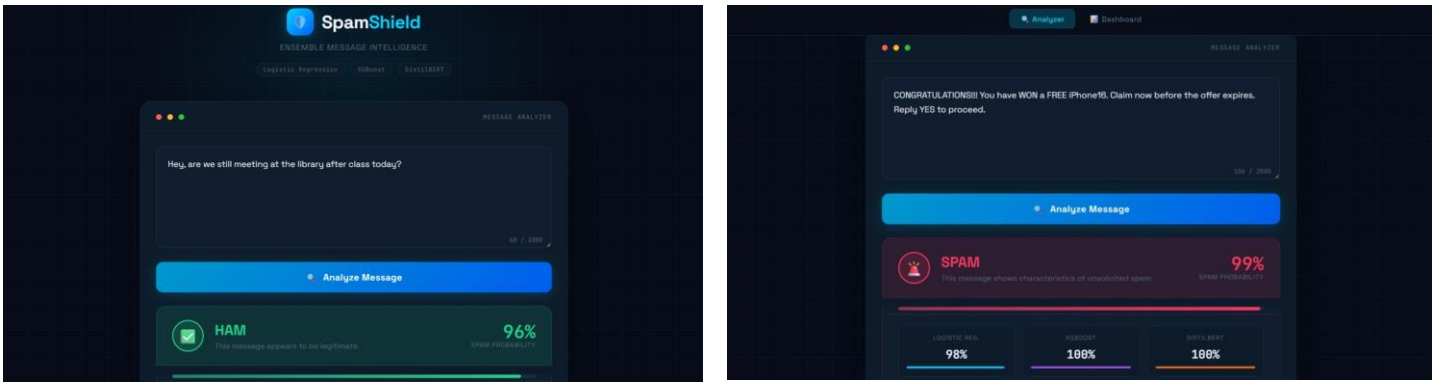


Figure 2. SMS Classification Result

Figure 2 demonstrates the output prediction made by the proposed system. Through the analysis of the SMS message using Logistic Regression, XGBoost, and DistilBERT. The ensemble model comes up with the prediction of either the SMS message being spam or ham. Besides the prediction result, the system provides the confidence level of the prediction made.



Figure 3. Model Statistics and Visualization Dashboard

In order to increase the ease of use and interpretation of the system, a dashboard was designed to displayed some of the critical statistics related to the SMS spam detection. This is shown in Figure 3. Some information displayed on the dashboard includes the performance of the models, the classification report, and confusion matrices.